I/Q imbalance predistortion schemes for OFDM systems

Jihoon Lee and Jin Kwan Kim

***Abstract*—In this paper, we propose two novel predistortion schemes to mitigate I/Q imbalance in Orthogonal Frequency Division Multiplexing (OFDM) communication systems: RF signal-based and deep learning-based schemes. OFDM serves as the baseline waveform for both current 5G and the emerging 6G wireless systems. With the RF signal-based scheme, gain and phase errors are estimated in real time using feedback from the RF signal. With the deep learning-based scheme, I/Q imbalance predistortion is performed using Bi-directional Long Short-Term Memory (BLSTM). The simulation results show that the proposed schemes improve the bit error probability in OFDM systems.**

***Index Terms*—I/Q imbalance, predistortion, deep learning, OFDM**

I. Introduction

With the evolution from 5G New Radio (NR) to the emerging sixth generation (6G) wireless systems, there is an increasing demand for higher data rates, lower latency, and massive connectivity. Both 5G and 6G systems adopt Orthogonal Frequency Division Multiplexing (OFDM) based waveform as their fundamental air interface [1], [2], [3]. Recent 3GPP standardization activities have reached a consensus that 6G will employ CP-OFDM for the downlink and DFT-s-OFDM for the uplink as the baseline waveforms [2], [3].

OFDM offers several advantages, including high spectral efficiency and strong robustness against multi-path fading and Doppler effects. It is also resilient to narrow-band interference and enables high-speed signal processing through Fast Fourier Transform (FFT) techniques, facilitating efficient hardware implementation. Moreover, OFDM systems do not require complex equalizers and shows strong resistance to impulse noise. Due to these characteristics, OFDM remains a widely adopted modulation scheme for ultra-high speed wired and wireless communication systems worldwide, with extensive ongoing standardization efforts based on OFDM technologies. However, OFDM systems are highly sensitive to impairments such as frequency offset, high Peak-to-Average Power Ratio (PAPR), DC offset, in-phase/quadrature imbalance, and phase noise [4], [5].

Modern wireless systems demand highly integrated, low-cost receivers. Zero Intermediate Frequency (zero-IF) receivers have been widely adopted [6]. However, while baseband processing is digital, up/down-conversion occurs in the analog domain, where I/Q imbalance is inevitable due to gain and phase mismatches. In addition, as wireless systems move toward higher frequency bands and wider bandwidths in 6G, the impact of I/Q imbalance becomes more severe, making effective compensation techniques increasingly critical.

Several researches have addressed I/Q imbalance [7], [8], [9], however these require pilot signals and focus on post-compensation. Since post-compensation results in larger residual errors, pre-compensation is preferable. We propose two predistortion schemes: (1) RF signal-based without pilots, and (2) BLSTM-based [10], [11], [12].

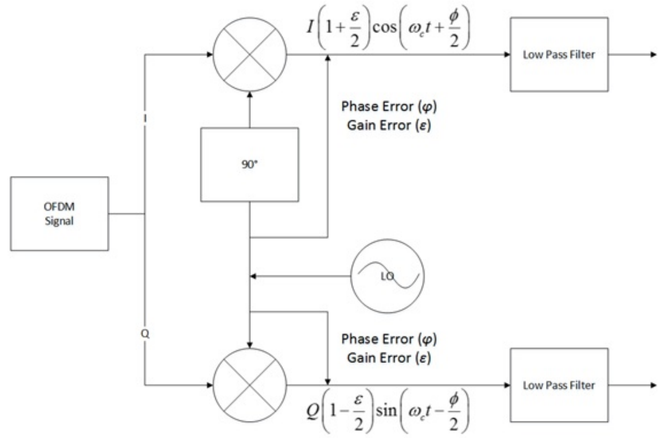

Fig. 1. I/Q imbalance of the local oscillator

The simulation results demonstrate that the proposed predistortion schemes improve the BER performance of the OFDM system under I/Q imbalance conditions.

II. System model

A. OFDM System

In an OFDM system, a high-rate data stream is divided into multiple low-rate streams. The time-domain OFDM signal $x(t)$ is

$$x(t) = \frac{1}{N}\sum_{n=0}^{N-1} X_n e^{j2\pi\Delta f t}, \quad 0 \le t < NT \tag{1}$$

where N is the number of subcarriers, X_n is the data symbol, and $\Delta f = 1/(NT)$ is the subcarrier spacing. In (1), the time-domain OFDM signals $x(t)$ can be expressed as a sum of in-phase signal $I(t)$ and quadrature signal $Q(t)$ ($x(t) = I(t) + jQ(t)$).

B. I/Q Imbalance

In the up-conversion process, in-phase and quadrature signals are mapped onto cosine and sine waves of Local Oscillator (LO), respectively. In ideal up-conversion, the RF signal $s(t)$ is

$$s(t) = I(t)\cdot cos(\omega_c t) + Q(t)\cdot sin(\omega_c t), \tag{2}$$

where ω_c is the carrier frequency of the RF signal.

Fig. 1 shows the I/Q imbalance of the LO. The signal with I/Q imbalance $\hat{s}(t)$ can be expressed as

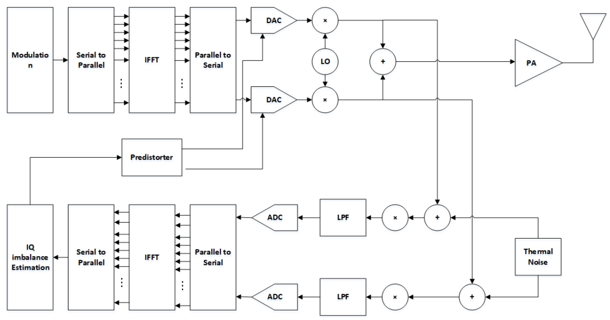

Fig. 2. Transmitter structure with I/Q imbalance predistorter based on RF signal.

$$\begin{aligned}\hat{s}(t) = I(t)\left(1+\frac{\varepsilon}{2}\right) cos\left(\omega_c t+\frac{\varphi}{2}\right)\\ + Q(t)\left(1-\frac{\varepsilon}{2}\right) sin\left(\omega_c t-\frac{\varphi}{2}\right),\end{aligned} \quad (3)$$

where ε and φ are the gain and phase errors, respectively.

III. I/Q IMBALANCE PREDISTORTION SCHEMES

A. RF signal based predistortion of I/Q imbalance

Fig. 2 shows the transmitter structure with I/Q imbalance predistorter based on the RF signal. The RF-based scheme uses feedback from the transmitted RF signal. A transmitter down-converts the RF signal using the LO used in the up-conversion process. The down-conversion signal $\tilde{x}(t)$ can be expressed as the sum of in-phase signal $\tilde{I}(t)$ and quadrature $\tilde{Q}(t)$ ($\tilde{x}(t) = \tilde{I}(t) + j\tilde{Q}(t)$). Then, $\tilde{I}(t)$ and $\tilde{Q}(t)$ can be expressed as

$$\begin{aligned}\tilde{I}(t) &= \int_0^t \hat{s}(\tau)\left(1+\frac{\varepsilon}{2}\right) cos\left(w_c\tau+\frac{\varphi}{2}\right) + n(\tau)d\tau\\ &= I_L(t)\left(1+\frac{\varepsilon}{2}\right)^2 - Q_L(t)\left(1-\frac{\varepsilon^2}{4}\right) sin\varphi + n_L(t),\end{aligned} \quad (4)$$

$$\begin{aligned}\tilde{Q}(t) &= \int \hat{s}(t)\left(1-\frac{\varepsilon}{2}\right) sin\left(w_c t-\frac{\varphi}{2}\right) + n(t)dt\\ &= Q_L(t)\left(1-\frac{\varepsilon}{2}\right)^2 - I_L(t)\left(1-\frac{\varepsilon^2}{4}\right) sin\varphi + n_L(t).\end{aligned} \quad (5)$$

where $I_L(t)$, $Q_L(t)$ and $n_L(t)$ represent the I and Q signals and internal thermal noise after Low Pass filtering, respectively.

From (4) and (5), by neglecting ε^2 terms (valid for $\varepsilon < 0.1$) and rearranging the equations to isolate the error terms, the gain error ε_T and the phase error φ_T can be estimated as

$$\varepsilon_T = \frac{I(t)\tilde{I}(t) - Q(t)\tilde{Q}(t) - (I^2(t) - Q^2(t))}{I^2(t) + Q^2(t)}, \quad (6)$$

$$\varphi_T = sin^{-1}\left(\frac{2I(t)Q(t) - (I(t)\tilde{Q}(t) - Q(t)\tilde{I}(t))}{I^2(t) + Q^2(t)}\right). \quad (7)$$

Applying these estimates reduces I/Q imbalance effects.

B. Deep learning based predistortion of I/Q imbalance

Under ideal conditions, I/Q imbalance predistortion scheme based on RF signals shows the BER performance similar to the case of no I/Q imbalance. However, the performance degradation of predistortion can be observed in environments with noise in the internal circuitry. This performance degradation is considered to result from the process of converting the RF signal through the DAC and the subsequent internal feedback loop involved in the RF signal-based predistortion scheme. In this subsection, the deep learning based predistortion of I/Q imbalance is proposed, which does not involve the internal feedback or DAC processes.

In this paper, we introduce BLSTM, which is one of the deep learning algorithms. Based on BLSTM, we propose a novel I/Q imbalance predistortion technique.

Unlike traditional LSTM, BLSTM processes the information in both the forward and backward directions (Bi-directions), considering both past and future patterns. When processing the data x_t at the current time t, the model can simultaneously analyze the influence of past data $(x_{t-1}, x_{t-2}, ...)$ and future data $(x_{t+1}, x_{t+2}, ...)$ on the current data x_t [11], [12], [13].

At the input layer, the real and imaginary parts of the complex signal affected by I/Q imbalance are separately fed into the BLSTM model. These signals pass through the both forward and backward directions of BLSTM, which learn time-dependent features. The forward and backward hidden states, denoted by $\overrightarrow{h}_t$ and $\overleftarrow{h}_t$, are expressed as

$$\overrightarrow{h}_t = LSTM_{forward}(x_t, \overrightarrow{h}_{t-1}), \quad (8)$$

$$\overleftarrow{h}_t = LSTM_{backward}(x_t, \overleftarrow{h}_{t+1}). \quad (9)$$

Then, the combined hidden state h_t, obtained by combining the forward and backward hidden states, can be expressed as

$$h_t = [\overrightarrow{h}_t, \overleftarrow{h}_t]. \quad (10)$$

The training of the BLSTM model is performed as follows: First, the pairs of signals, each consisting of a signal with I/Q imbalance and its corresponding ideal target signal, are prepared to form the training dataset. Second, the loss function L is defined as the Mean Squared Error (MSE) between the ideal target signal and the compensated signal. Then, the loss function can be expressed as

$$L = \frac{1}{N}\sum_{i=1}^{N}(|I_{i,ideal} - I_{i,comp}|^2 + |Q_{i,ideal} - Q_{i,comp}|^2), \quad (11)$$

where $I_{i,org}$ and $Q_{i,org}$ are in-phase and quadrature part of the originally predistorted signal, respectively. On the other hand, $I_{i,comp}$ and $Q_{i,comp}$ are in-phase and quadrature part of the predistorted signal obtained through training, respectively. Finally, the model is trained using the training dataset and the defined loss function. Through the optimization process, the model learns to effectively compensate I/Q imbalance by minimizing the error between the ideal and the compensated signals.

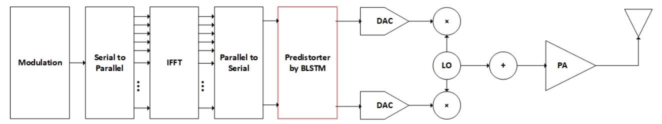

Fig. 3. The example of transmitter structure with I/Q imbalance predistorter by BLSTM

For I/Q imbalance predistortion, the use of BLSTM algorithm offers several advantages. It can effectively compensate non-linear I/Q imbalance, which is challenging for conventional linear compensation algorithms. Moreover, BLSTM is capable of capturing the time-dependent changes in I/Q imbalance, making it suitable for dynamically changing environments. However, the BLSTM-based I/Q imbalance predistortion also presents certain limitations. It requires a large amount of training data to achieve reliable performance, and the training process can be computationally intensive and time-consuming. In addition, the implementation may demand more computational resources compared to traditional compensation techniques.

The BLSTM model consists of an input layer, two stacked BLSTM hidden layers (with 70 and 90 units in each direction, respectively), and an output layer that generates 32 compensated I/Q values. The first layer captures fundamental I/Q imbalance patterns across 32 subcarriers, while the second layer models non-linear interactions. This achieves near-optimal MSE performance while maintaining processing within one OFDM symbol duration. To train the BLSTM model, the Root Mean Square Propagation (RSMProp) optimizer was used. The batch size was set to 32, and training was conducted over 60 epochs. The training dataset consisted of a total 3,200,00 OFDM symbols, which were split into training (2,560,000 symbols) and test (640,000 symbols) sets at a ratio of 4:1. The performance of the training model was evaluated using the MSE. On the test set, the model achieved an MSE of 2×10^{-4}.

In particular, the model demonstrated stable predistortion performance even for I/Q imbalance parameters that were not included in the training data, validating its strong generalization capability. These experimental results confirm that the proposed BLSTM architecture is effective for I/Q imbalance predistortion and achieves the performance suitable for practical system implementation.

Fig. 3 shows the example of OFDM transmitter structure including the BLSTM used for I/Q imbalance predistortion. In Fig. 3, the BLSTM is applied after the Parallel-to-Serial (P/S) conversion. During the BLSTM processing, the OFDM symbols are predistorted to mitigate the I/Q imbalance and then up-converted to the RF signal.

IV. Simulation Results

In this section, simulation results are presented. The OFDM signal is modulated by Quadrature Phase Shift Keying (QPSK). The number of subcarriers is set to 32. For BLSTM training, the input and target data are set to signal with I/Q imbalance and the corresponding ideal signals, respectively. The BLSTM training is performed independently for the I

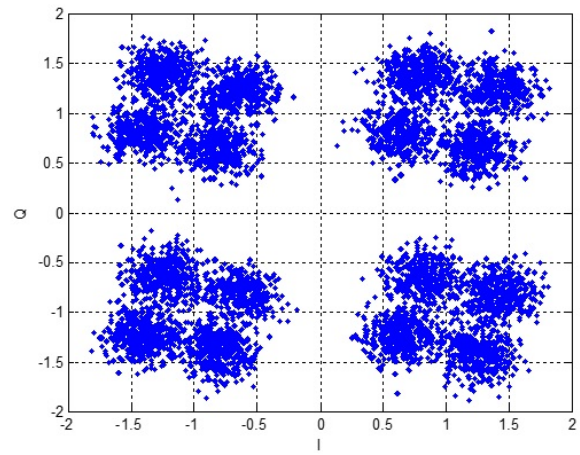

Fig. 4. Signal constellation in an AWGN channel with both amplitude and phase imbalances

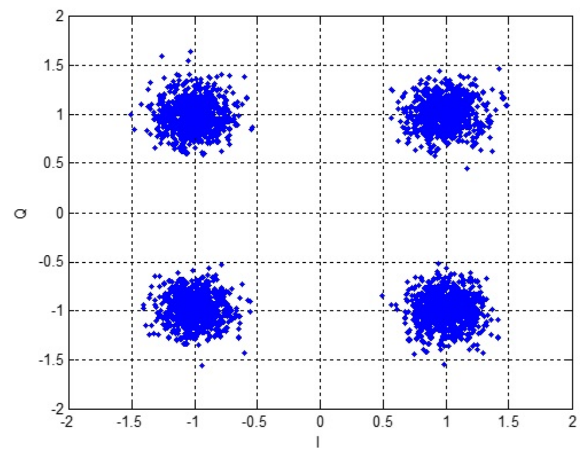

Fig. 5. Signal constellation in an AWGN channel after I/Q imbalance compensation

and Q components. In this paper, based on the values used in preliminary research, the number of hidden units in BLSTM are set to 70 and 90 for the first and second layers, respectively.

A. *Signal constellation under I/Q imbalance*

Figs. 4 and 5 show the signal constellations in an AWGN channel that set to E_b/N_0 of 15dB with I/Q imbalance and after compensation of the I/Q imbalance using BLSTM based predistortion, respectively. The constellation points become dispersed when either amplitude or phase imbalance is present individually. However, the constellation is both dispersed and rotated when both imbalances are present, as shown in Fig. 4. Compared to other RF impairments such as phase noise and non-linear amplification, I/Q imbalance is typically modeled with discrete amplitude and phase error parameters. This imbalance not only causes rotation of the constellation but also leads to the formation of multiple district clusters, rather than a simple spreading of points. From the results in Figs. 4 and 5, it can be observed that the constellation cloud is

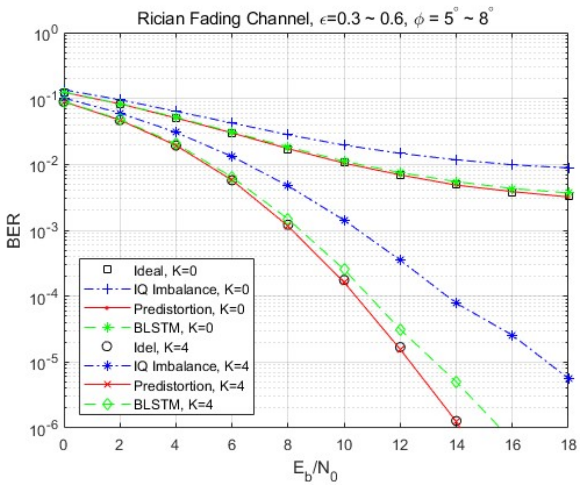

Fig. 6. BER performance with and without I/Q imbalance predistortion in Rician fading channel

significantly reduced after the compensation of I/Q imbalance using BLSTM based predistortion.

B. BER performance

Fig. 6 shows the BER performance of two proposed predistortion schemes in an Rician channel with K factors of 0 and 4. For comparison, the results for the uncompensated case and the ideal case (no I/Q imbalance) are also plotted. The BLSTM based predistortion model was trained on signal with I/Q imbalance parameters of $\epsilon = 0.3 \sim 0.6$ and $\varphi = 5 \sim 8°$. For the BER of 10^{-2}, the RF signal based and BLSTM based predistortion schemes achieve approximately 6.0 dB and 5.5 dB performance improvements, respectively, compared to the uncompensated case under a Rician fading channel with a K factor of 0. For the BER of 10^{-5}, the RF signal based and BLSTM based predistortion schemes achieve approximately 4.5 dB and 4.0 dB performance improvements, respectively, compared to the uncompensated case under a Rician fading channel with a K factor of 4.

The RF signal-based scheme shows near-ideal performance. The BLSTM-based scheme exhibits about 0.5dB performance gap because the RF scheme directly measures errors in real time, while BLSTM relies on training data covering limited ranges. Moreover, the presence of residual MSE error (1.2×10^{-4}) and the computational complexity constraints that limit the model capacity result in the 0.5 dB performance gap compared to the RF signal-based scheme. However, the performance of the BLSTM-based predistortion scheme could be further improved by using more refined training data and the application of optimal layers and parameters.

V. Conclusion

This paper proposed two novel predistortion schemes for I/Q imbalance mitigation in OFDM systems. These two predistortion schemes overcome the limitations of conventional compensation techniques that require additional pilot or reference signals, and provide an effective solution for mitigating I/Q imbalance with non-linear characteristics.

The performance of the proposed predistortion schemes is evaluated by simulation. The simulation results show that the RF signal based predistortion scheme achieves performance nearly identical to the ideal case without I/Q imbalance, whereas the BLSTM-based predistortion scheme shows a performance degradation of about 0.5 dB. This degradation is mainly due to the limitations of this paper, including the need for sufficient training data and the considerable time required for initial training. With further improvements in training strategies and model capacity, the BLSTM-based predistortion scheme has the potential to reach performance comparable to that of the RF signal based predistortion scheme.

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